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1- REQUIRED TOOLS

- Nonferrous flat-head screwdriver
- Nonferrous hexagon (Allen) wrench set
- Regular screwdriver
- Nonferrous wrenches

Description - Replacement procedures for hardware located in the rear pedestal. For non-Cx systems refer to procedure for Rear Pedestal Hardware Replacement for Non-Cx Magnet, RP1REA1.PDF.

2- LONGITUDINAL ENCODER

The Longitudinal Encoder (MG3A6), shown in Illustration 2-1, is located in the rear pedestal (MG3).

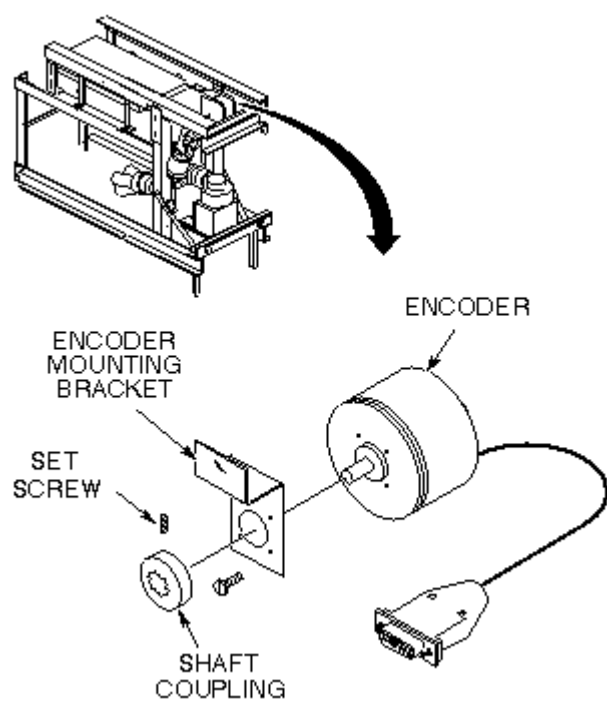


ILLUSTRATION 2-1
LONGITUDINAL ENCODER LOCATION

2-1 Removing Old Encoder

1. Switch off circuit breaker MR1 A14 CB6 at rear of RF/Pen Cabinet (MR1) on Magnet Enclosure Power Supply (MEPS) module.
2. Lock-Out / Tag-Out MR1 A14 CB6 using Safety procedure, SYSSMA2.PDF.
3. Verify that voltage between MR1 A14 TP1 (+) and TP14 (-) is **0 VDC**.
4. Remove rear pedestal right side cover (as viewed from rear of magnet).
5. Disconnect encoder cable from J3 of Magnet I/F #1 (MG3 A1).
6. Loosen set screw on shaft coupling. (See Illustration 2-1.)



FERROUS MATERIAL HAZARD!! ENCODER CONTAINS FERROUS MATERIAL. HOLDING THIS ASSEMBLY TOO CLOSE TO THE MAGNET BORE WILL FORCIBLY ATTRACT IT TO THE MAGNET. TO PREVENT POSSIBLE BODILY INJURY OR MATERIAL DAMAGE, REMOVE ENCODER AND MOUNTING BRACKET FROM MAGNET ROOM TO INSTALL NEW ENCODER.

7. Remove screw from encoder mounting bracket, and remove encoder and mounting bracket from rear pedestal. (See Illustration 2-1.) Remove encoder and mounting bracket from the Magnet Room.
8. Remove the four screws that hold encoder to mounting bracket.

2-2 Installing Replacement Encoder

1. Install new encoder onto mounting bracket with four screws.
2. Install bracket and encoder assembly onto rear pedestal.
3. Tighten set screw on shaft coupling.
4. Connect encoder cable to J3 of Magnet I/F #1.
5. Replace rear pedestal right side cover.
6. Remove Lock-Out / Tag-Out. Switch on circuit breaker MR1 A14 CB6 at rear of RF/Pen Cabinet (MR1) on MEPS module.
7. Adjust encoder voltage. Refer to the procedure for *Longitudinal Drive System Calibration, Longitudinal Encoder Supply Voltage Adjustment* (PT1SCA1.DOC).
8. Starting with cradle at Home position, drive table into bore until longitudinal position display reads **10 mm**.
9. Place tape marker on cradle and adjacent spot on table.
10. Drive table into bore until longitudinal position display reads **1010 mm**.
11. Measure distance between marks and verify that distance traveled is 1000 mm, ± 1 mm.

3- LONGITUDINAL DRIVE MOTOR REPLACEMENT

The Longitudinal Drive (MG3A7) is located in the rear pedestal. Two people are required to replace the Longitudinal Drive Motor due to its strong attraction to the magnet.

1. Perform Lock-Out / Tag-Out of the PDU as described in the Safety procedure, [SYSMSA2.PDF](#).
2. Remove rear pedestal left side cover (as viewed from rear of magnet).
3. Disconnect wires from terminals on motor.
4. Loosen lower set screw on motor shaft coupling.



FERROUS MATERIAL HAZARD! THE MOTOR IS MADE OF FERROUS MATERIAL. HOLDING THIS ASSEMBLY TOO CLOSE TO THE MAGNET BORE WILL FORCIBLY ATTRACT IT TO THE MAGNET. TO PREVENT POSSIBLE BODILY INJURY OR DAMAGE TO MOTOR, MAGNET ENCLOSURE OR MAGNET, DO NOT LIFT THIS MOTOR ABOVE THE LEVEL OF THE MOUNTING ON THE REAR PEDESTAL FRAME. TWO PEOPLE ARE REQUIRED FOR THIS PROCEDURE.

5. Refer to Illustration 3-1 for the steps that follow. With a second person holding onto the motor firmly with both hands, loosen and eventually remove the two long screws holding the assembly in place. (While removing the two long motor mounting screws from the bracket, the aluminum spacers will become free as well.)
 - a. When the last screw has been removed, both people should maintain control of the longitudinal drive motor and carefully move it to the wall on the coldhead side of the magnet. One person may slide the longitudinal drive motor along the floor on the cold head side of the magnet and exit the Magnet Room.
 - b. Retain the two long screws and aluminum spacers for installation of replacement motor.

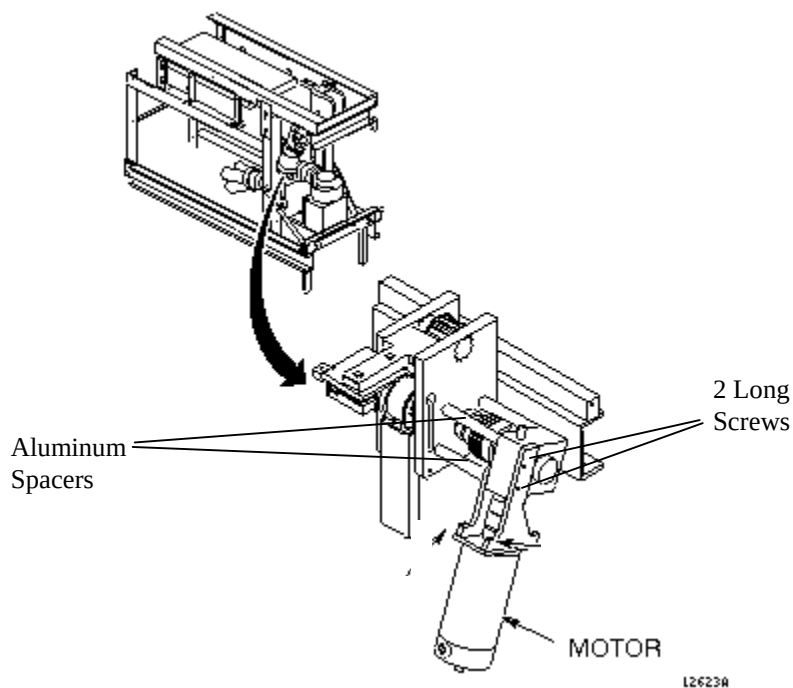


ILLUSTRATION 3-1
LONGITUDINAL DRIVE MOTOR REMOVAL

6. With the motor OUTSIDE of the Magnet Room, remove the 4 mounting screws from the motor mounting bracket. These screws go through the mount and secure the motor by pulling it tight up to the mount. (Refer to Illustration 3-2.)

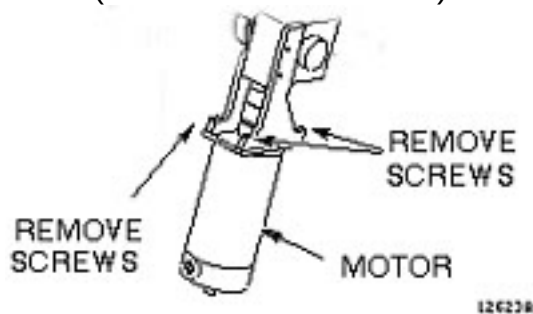


ILLUSTRATION 3-2
MOTOR AND BRACKET SEPERATION

7. Install the replacement motor on to the bracket. Re-install the 4 mounting screws. Draw the 4 mounting screws up evenly tightening the motor in its mount. (See Illustration 3-2.)
8. One person may slide the longitudinal drive motor assembly along the floor on the coldhead side of the magnet to the location of the rear pedestal.
 - a. With two people firmly holding the motor assembly, lift the motor assembly from the floor and reinstall it into the rear pedestal.
 - b. With one person holding onto the motor firmly with both hands, reconnect the motor and bracket assembly to the clutch assembly with the two long screws and aluminum spacers.
9. Tighten the screw on the coupling to finish the motor installation.
10. Re-attach the electrical cable to the motor.
11. Refer to the Safety procedure, SYSMSA2.PDF. Remove Lock-Out / Tag-Out.

4- HOME LIMIT SWITCH BOARD REPLACEMENT

The Home Limit Switch Board (MG3 A1 A1) is located in the rear pedestal inside of the Magnet I/F #1 (MG3 A1) shown in Illustration 4-1.

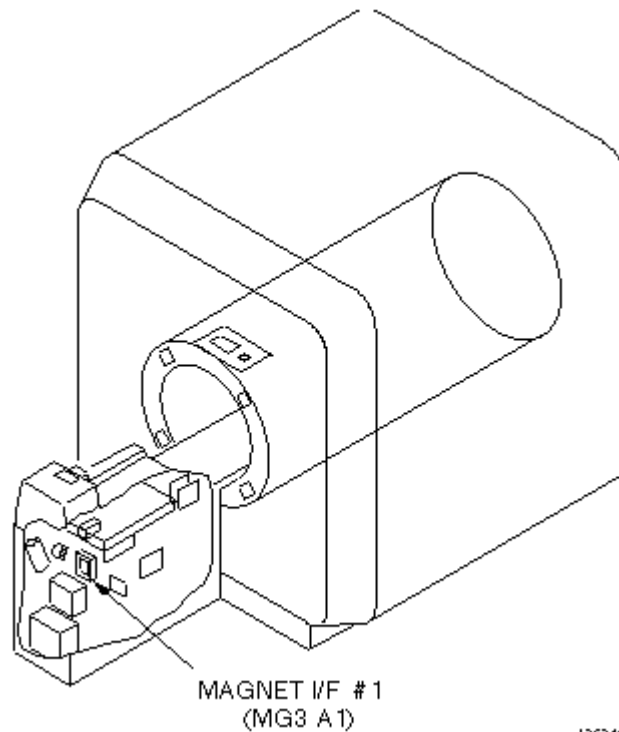


ILLUSTRATION 4-1
LOCATION OF MAGNET I/F #1

1. Switch off MR1 A14 CB6 at rear of RF/Pen Cabinet (MR1) on magnet enclosure power supply (MEPS) module.
2. Lock-Out / Tag-Out MR1 A14 CB6 using Safety procedure, SYSMSA2.PDF.
3. Verify that voltage between MR1 A14 TP1 (+) and TP14 (–) is **0 VDC**.
4. Remove rear pedestal right side cover (as viewed from rear of magnet).
5. Remove screws from cover of Magnet I/F #1.
6. Disconnect sensor plug from J2 of Home Limit Switch Board.
7. Disconnect harness from J3 of Home Limit Switch Board.
8. Replace failed board.
9. Connect harness to J3 of Home Limit Switch Board.
10. Connect sensor plug from J2 of Home Limit Switch Board.
11. Install screws onto cover of Magnet I/F #1.

12. Install rear pedestal right side cover.
13. Remove Lock-Out / Tag-Out from circuit breaker MR1 A14 CB6 at rear of RF/Pen Cabinet on MEPS module.
14. Switch on circuit breaker MR1 A14 CB6.
15. Verify that cradle returns to Home position and longitudinal drive motor has stopped trying to drive the cradle.

5- REAR PEDESTAL AIR HOSE

Replace rear pedestal air hoses per Illustrations 5-1 and 5-2.

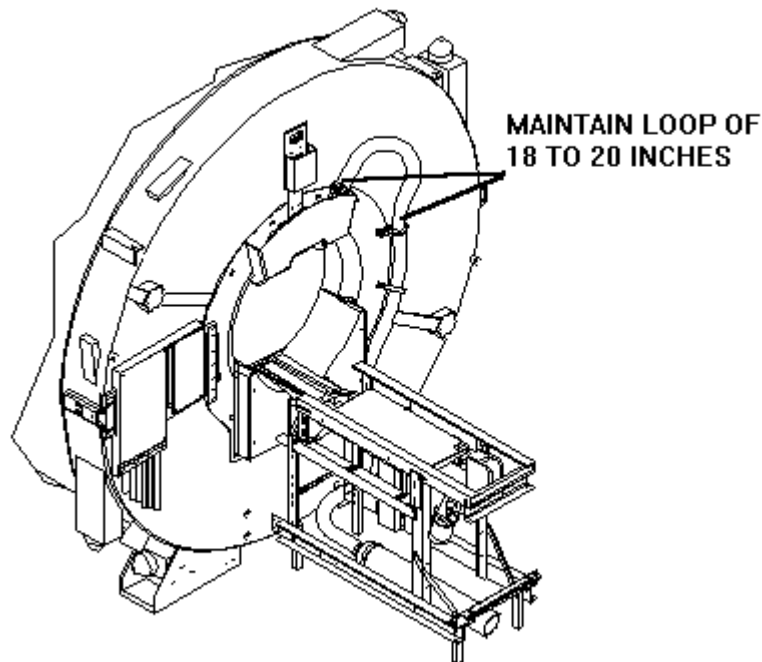


ILLUSTRATION 5-1
CX AIR HOSE ROUTING

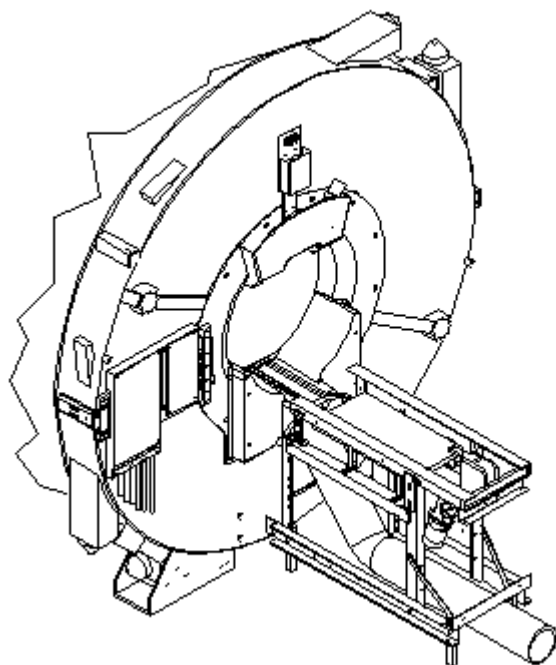


ILLUSTRATION 5-2
CX AIR HOSE ROUTING

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	Aug 19, 1998	R. Hawthorne	Initial conversion to Word
1	May 21, 1999	SM Atladottir	Updated Procedure References for the New GUI
2	Sept. 6, 2001	D. Thome	Updated Long. Drive Motor replacement process.
3	Sept. 30, 2010	R. Kohl	Per CAPA 1859293: Added text to Steps 5 and 8 in Section 3 - Longitudinal Drive Motor Replacement.